and the total number of live births as

$B_t$ = 4,130.

Hence, the Crude Birth Rate of the town is

$$CBR = \frac{4130}{189000} \times 1000$$

= 21.85 per thousand.

*CBR* is the simplest fertility rate, which provides a comprehensive idea about the population growth of any region/community. It is easy to compute for any given region.

However, *CBR* does not take into account of the age and gender distribution of the population or specifically the number of women at the child bearing age. This is a crude measure, since it includes the sections of the population who are not in the reproductive age group. It means that $P_t$ includes the sections of the population who are not exposed to the risk of producing children, in particular, male and also the female beyond the reproductive age. *CBR* also assumes that all the women at child bearing age have the same reproductive capacity irrespective of their age, which is totally unrealistic.

### (ii) General Fertility Rate

**General Fertility Rate** (*GFR*) of a region or a community relates the number of live births to the number of women in the reproductive age.

*i.e.*,
$$GFR = \frac{Number\ of\ \ live\ births}{Number\ of\ women\ at\ child\ bearing\ age} \times 1000$$

This quantity can be computed using the formula

$$GFR = \frac{B_t}{\sum_{i=a_1}^{a_2} P_t^i} \times 1000$$

where

$P_t^i$ denotes the number of women in the reproductive age $i$ years in the given region/community during the period $t$, $i = a_1$ to $a_2$.

In India, generally, $a_1$ = 15 years and $a_2$ = 49 years.

*GFR* overcomes the disadvantage of *CBR* considering only the women population at the child bearing age group, since the denominator in the above formula represents the entire women population at the reproductive age group. *GFR* expresses the increase in the women population at the child bearing age through live births.

However, *GFR* does not express the age composition of women population at the reproductive age group. Hence, two different regions/communities cannot be compared with respect to age of women using *GFR*.



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